E-Learning Platform Purchase Analysis using MySQL

1. Executive Summary

This project analyses learner purchases on an online learning platform using MySQL and Tableau. The objective was to study learner spending behaviour, identify top-performing courses and categories, analyze cross-category purchasing patterns, and detect courses with low adoption. SQL was used for relational modelling and analytical querying, while Tableau was used to visualize sales trends and learner insights interactively.

2. Business Problem

Online learning platforms need to understand which courses generate the highest revenue, which learners contribute most to sales, and which categories drive engagement. Without proper analysis, business teams may struggle to optimize pricing, marketing, and course recommendations.

3. Project Objective

- Analyze learner purchase behaviour
- Calculate learner-wise spending
- Identify top-selling courses
- Evaluate category-wise revenue
- Detect multi-category learner behaviour
- Identify courses with zero purchases
- Build an interactive Tableau dashboard

4. Dataset Overview

- 88 Learners
- 6 Courses
- 80+ Purchase transactions

5. Limitations:

• Dataset size is simulated and limited compared to real enterprise systems.

• Purchase trends are based on sample historical data.

• No real-time streaming or predictive modelling was implemented.

6. E-Learning Dataset Attributes

Table Name	Attribute	Description
Learners	learner_id	Unique learner ID
	full_name	Learner name
	country	Learner country
Courses	course_id	Unique course ID

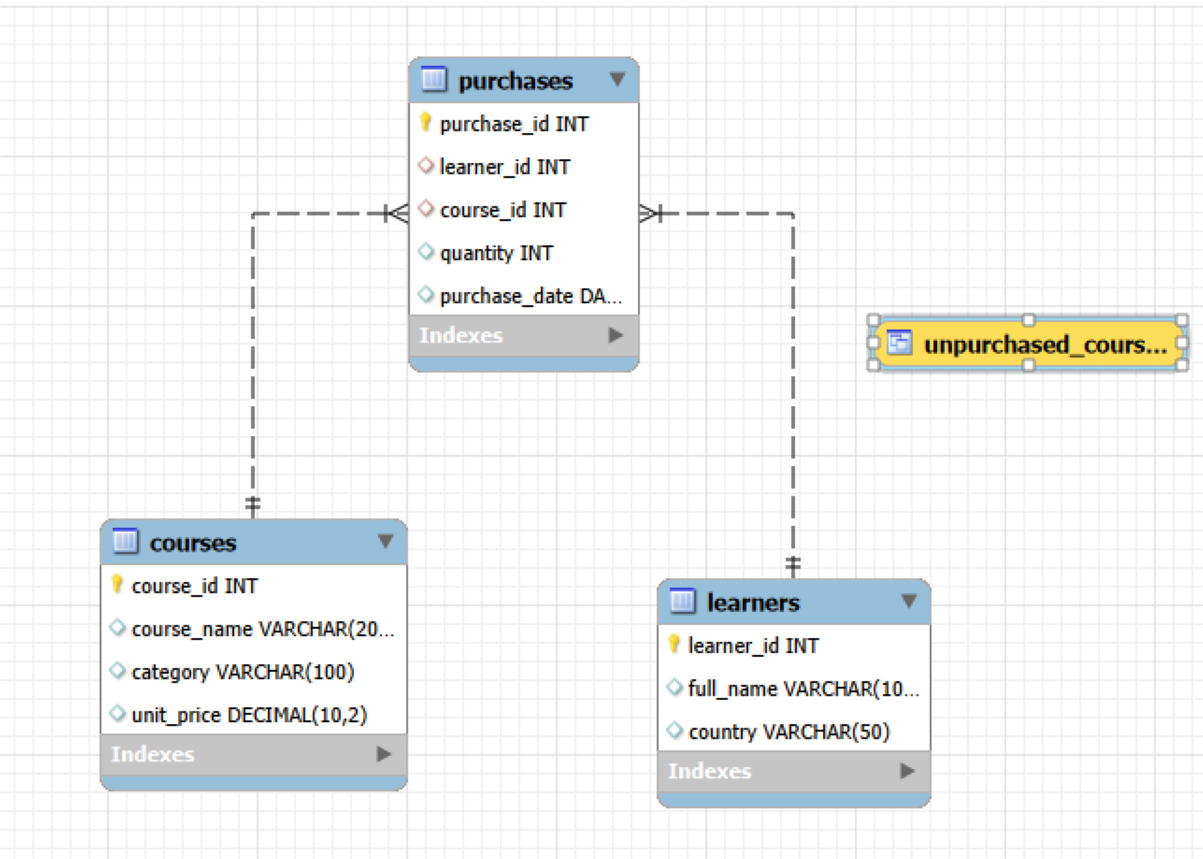
	course_name	Course title
	category	Course category
	unit_price	Course price
Purchases	purchase_id	Unique purchase ID
	learner_id	Purchased learner
	course_id	Purchased course
	quantity	Number of purchases
	purchase_date	Purchase date

7. Tools & Technologies Used

- MySQL
- MySQL Workbench
- Tableau
- SQL
- Relational Database Design

8. Database Design

Figure1: EER Diagram



The database follows a relational schema where the purchases table acts as the transactional fact table. The learners and courses tables function as dimension tables linked through foreign key relationships. This design enables efficient analytical querying and aggregation.

9. SQL Analysis Performed

The analysis was performed using the following concepts

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- GROUP BY
- HAVING
- ORDER BY
- LIMIT
- Aggregate Functions
- CTE
- SQL Views
- Window Functions
- Multi-Table Relational Analysis
- Business Performance Analytics

10. Key Business Questions

- ➢ Which learners contribute the highest spending?
- ➢ Which courses generate the highest demand?
- ➢ Which categories generate maximum revenue?
- ➢ Which learners purchase across multiple categories?
- ➢ Which courses are underperforming?

11. SQL Query Logic

Database Creation & Data Modelling

The project began with the creation of a relational database schema consisting of learners, courses, and purchases tables. Primary and foreign key constraints were implemented to maintain referential integrity between transactional and dimensional data. Additional learner and purchase records were inserted to simulate realistic platform activity and improve analytical depth.

Relational Data Exploration using Joins

Different SQL joins were used to combine transactional and master datasets for business analysis:

- INNER JOIN was used to retrieve learner purchase transactions with course details.
- LEFT JOIN was used to identify learners without purchases.
- RIGHT JOIN was used to detect courses with no purchase activity.

These joins enabled complete relational analysis across learners, purchases, and courses tables.

Learner Spending Analysis using CTE

A Common Table Expression (CTE) was implemented to calculate total learner spending using:

SUM(quantity * unit_price)

The CTE simplified reusable analytical logic and improved query readability while aggregating learner-level revenue contribution. This analysis helped identify high-value learners and spending concentration patterns.

Category Revenue Analysis

Category-level revenue was calculated by aggregating purchase quantities and course pricing across all transactions. GROUP BY and aggregate functions were used to evaluate:

- total revenue by category,
- learner participation,
- and category contribution trends.

The analysis identified Business Intelligence as the leading revenue-generating category on the platform.

Course Performance Analysis

Window functions were used to analyze course demand without collapsing detailed records. The following analytical functions were implemented:

SUM() OVER(PARTITION BY)

DENSE_RANK()

These functions enabled ranking of top-performing courses based on total quantities sold while handling ranking ties effectively. Power BI Dashboarding and Tableau Storytelling emerged as the highest-performing courses.

Multi-Category Learner Behaviour Analysis

HAVING clauses and DISTINCT category aggregation were used to identify learners purchasing across multiple learning categories. This analysis helped detect:

- highly engaged learners,
- cross-category purchase behaviour,
- and potential upselling opportunities.

Country Distribution Analysis

Learner distribution across countries was analyzed using grouped aggregations on learner geographic data.

This helped identify:

- High-performing regional markets,

- Learner concentration,
- Country-level demand trends.

India represented the highest learner share among all countries.

Underperforming Course Detection

LEFT JOIN and NULL filtering techniques were used to identify courses with no purchase activity. This analysis helped evaluate low-engagement or underperforming courses requiring marketing or pricing optimization.

SQL Views for Dashboard Integration

SQL Views were created to simplify reusable analytical outputs and support Tableau dashboard integration. The views centralized KPI calculations and improved query maintainability for business reporting and visualization workflows.

Time-Based Purchase Trend Analysis

Purchase transactions were analyzed using purchase dates grouped by month to identify revenue fluctuations and learner engagement trends over time. This analysis revealed strong revenue growth during Month 3 followed by a significant decline in Month 4, highlighting changing platform engagement patterns.

Business Intelligence Reporting

The final SQL outputs were integrated into Tableau to create an interactive dashboard containing:

- KPI metrics,
- Revenue analysis,
- Learner segmentation,
- Country distribution,
- Top-selling courses,
- Purchase trend visualizations.

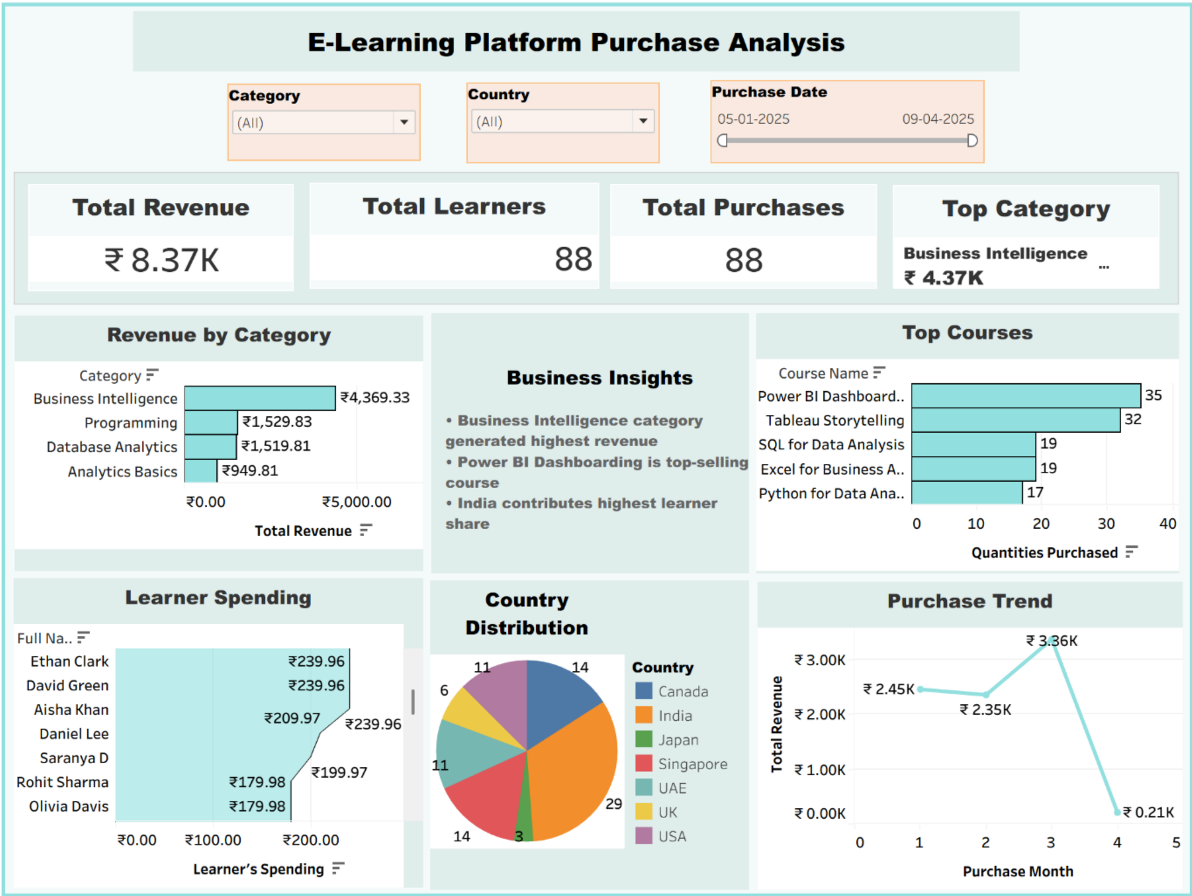
The combined SQL and Tableau workflow enabled end-to-end business intelligence reporting and decision-support analysis.

12. Tableau Dashboard Overview

The dashboard includes:

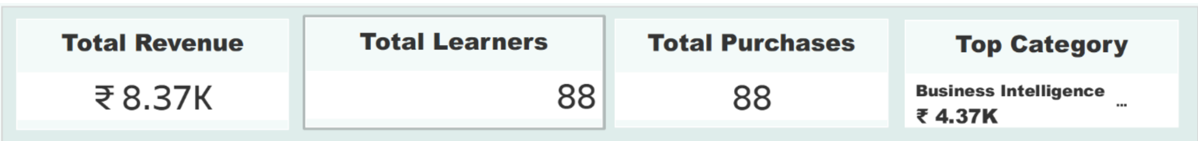
- KPI cards for revenue, learners, and purchases
- Revenue by category analysis
- Top-selling courses
- Country-level learner distribution
- Monthly purchase trend analysis

Figure 2 : Dashboard



13. KPI Summary:

Figure 3 : KPI Cards



14. Key Insights

1. Business Intelligence Dominates Platform Revenue:

The Business Intelligence category generated the highest platform revenue at approximately ₹4.37K, contributing more than 50% of the total revenue. This indicates strong learner demand for BI and dashboarding-related skills**.**

2. Power BI Dashboarding is the Top-Selling Course:

Power BI Dashboarding recorded the highest number of purchases (35), making it the most popular course on the platform. Tableau Storytelling followed closely with 32 purchases, showing high market interest in visualization and reporting skills.

3. India Represents the Largest Learner Base:

India contributed the highest learner share with 29 learners, significantly outperforming other countries. This suggests strong market potential for analytics-focused learning programs in the Indian region.

4. Revenue Concentration Exists Across Few Categories:

A major portion of platform revenue is concentrated within Business Intelligence and Programming categories, while Analytics Basics generated comparatively lower revenue. This indicates uneven category performance across the platform.

5. Revenue Peaked in Month 3 Before Sharp Decline:

Platform revenue increased steadily and peaked at ₹3.36K during Month 3 before dropping sharply in Month 4. This trend may indicate seasonal purchasing behaviour, reduced campaign activity, or limited new course engagement.

6. High-Spending Learners Contribute Significantly:

A small segment of learners contributed disproportionately higher spending compared to the overall learner base. This creates opportunities for premium subscriptions, certification programs, and advanced course upselling.

15. Business Recommendations

1. Expand High-Performing BI Courses:

Since Power BI Dashboarding and Tableau Storytelling show strong demand, the platform should introduce:

- Advanced BI Tracks,
- Real-World dashboard projects,
- Certification-based learning paths.

2. Strengthen Low-Revenue Categories:

Analytics Basics generated comparatively lower revenue despite strong industry relevance. Pricing optimization, promotional campaigns, or beginner-friendly bundles can improve adoption.

3. Focus Marketing on India:

Since India contributes the highest learner share, region-specific promotions and localized marketing campaigns can help increase conversions and retention.

4. Introduce Bundle Learning Paths:

Combining:

- SQL,
- Excel,
- Power BI,
- Python

into structured analyst career tracks can increase cross-course purchases and learner retention.

5. Improve Revenue Stability:

The sharp Month 4 decline suggests the need for
Recurring promotional campaigns,
Learner engagement emails,
Limited-time discounts,
Subscription-based offerings.

16. Conclusion

This project demonstrates how SQL and Tableau can be combined to analyze learner behaviour, sales performance, and category-level business trends in an e-learning environment. The project highlights the role of relational databases and data visualization in supporting business decision-making. The analysis demonstrates how data-driven insights can help e-learning businesses optimize course strategy, improve learner targeting, and maximize revenue generation.

17. Future Improvements

- Real-Time Sales Tracking with Larger Datasets
- Implement predictive models to forecast learner purchases and category demand.
- Learner Retention Analysis
- Cohort Analysis